

Raymond Liu

 [website](#) |  [LinkedIn](#) |  liur@seas.harvard.edu

EDUCATION

Harvard University

- PhD, Computer Science Expected May 2029
Co-advisors: Patrick Slade, Elena Glassman
- SM, Computer Science May 2026

Princeton University

- BSE, Computer Science May 2023
- Minor in Statistics and Machine Learning

PUBLICATIONS

“Improving outdoor navigation for people with blindness using an AI-driven smartphone application and personalized audio guidance”

Raymond Liu and Patrick Slade

Nature Biomedical Engineering, 2026 [\[Link\]](#)

“Training methods for human-robot interaction”

Raymond Liu, Ava Lakmazaheri, Estelle Richard, Louis Marchand, Elena Glassman, and Patrick Slade

In preparation

“Elbow-based MoE Routing: A Training-Free Inference Time Plugin for Expert Selection”

Robin Pan, Raymond Liu, Daniel Fang, Adelina Andrei, and Rosa Wa

ICLR Workshop on Test-Time Updates, 2026 (Accepted oral presentation) [\[Link\]](#)

WORK EXPERIENCE

Visiting Researcher at Harvard Slade Lab

July 2023 – Aug. 2024

- Developing a smartphone-based navigation assistant for people who are blind or visually impaired.
- For this work, we received funding from the Amazon Greater Boston Tech Initiative (\$50k) and the Harvard GRID Accelerator (\$100k).

Software Engineering Intern at Amazon

May 2022 – Aug. 2022

- Developed, tested, and optimized an API in Java to send parallelized batch requests to Elasticsearch and greatly speed up workflows.

TALKS, PANELS, AND EXHIBITS

[E3] Tech demo at *Abilities: Amplified*, Hatch Memorial Shell, Boston, MA, June 2026. [\[Event page\]](#)

[T1] Raymond Liu. “Smartphone Navigation Assistant for People with Impaired Vision,” *Harvard Robotics Trainee Seminar Series*, Harvard University, Aug. 2025.

[E2] Exhibitor at *DisabilityTech: Innovation for an Accessible Future*, Museum of Science, Boston, MA, Aug. 2025.

[P2] Raymond Liu, Patrick Slade, Erich Manser. “Accessibility Drives Innovation.” Panel at *Harvard Accessibility Summit*, Harvard University, June 2025. [\[Event page\]](#)

[P1] Panel at *Revolutionizing Life: Frontiers of Biotechnology*, Harvard OpenBio Lab, June 2025. [\[Video\]](#)

[E1] Raymond Liu. “A Personalized Navigation Assistant for People with Impaired Vision or Blindness.” Poster with demo at *Northeast Robotics Colloquium (NERC)*, UMass Amherst, Sep. 2024. [\[Event page\]](#)

TEACHING AND OUTREACH

Teaching Fellow at Harvard

Sep. 2025 – May 2026

- ES224/BE124 – Biomechanics of Movement and Assistive Robotics
- Gave guest lectures on personalizing assistive devices in the real world.
- Guided students during regular office hours; graded and provided feedback for assignments and exams.
- Course content: biomechanical principles of movement, simulation with musculoskeletal modeling, and topics in assistive robotics.

National Biomechanics Day at Harvard

May 2024 – May 2026

- Helped host annual [National Biomechanics Day](#) events at Harvard; demonstrated machine learning and wearable sensing concepts to Boston-area high school students.

Teaching Assistant at Princeton

Sep. 2021 – May 2023

- COS340 – [Reasoning about Computation](#)
- Guided students during regular office hours; graded and provided feedback for assignments and exams.
- Course content: mathematical and theoretical topics in computer science – combinatorics, probability, graph theory, NP-completeness, and cryptography.

RESEARCH MENTORING

Master's Students:

Ambre Jade Sassi Robotics and Neural Eng. (Visiting from EPFL) <i>Thesis</i>	Mar–Sep 2026
Louis Marchand Robotics and Data Science (Visiting from EPFL) <i>Thesis</i>	Mar–Sep 2025
Estelle Richard Robotics (Visiting from EPFL) <i>Thesis</i>	Mar–Sep 2024

Bachelor's Students:

Serena Liu Mechanical Engineering	Fall 2025
Ali Saffarini Electrical Engineering	Spr 2024, Fall 2024, Spr 2025
Alina Yu Computer Science	Fall 2024
Pooja Vegesna Computer Science (Visiting from UChicago)	Summer 2024
Nico Fidalgo Computer Science	Fall 2023
Kuma McCraw Mechanical Engineering	Fall 2023

HONORS AND AWARDS

NSF Graduate Research Fellowship Awardee

Apr. 2024

Harvard GRID Accelerator Awardee

Feb. 2024

- Received \$100k grant for our proposal, “Navigation assistant for people with impaired vision or blindness.”

Amazon Robotics Greater Boston Tech Initiative Awardee

Dec. 2023

- Received \$50k grant for our proposal, “Curriculum for developing a robotic navigation assistant for people with visual impairment.”

NSF Graduate Research Fellowship Honorable Mention

Mar. 2023

USA Junior Mathematical Olympiad Qualifier

Apr. 2018

- One of 156 qualifiers nationwide.